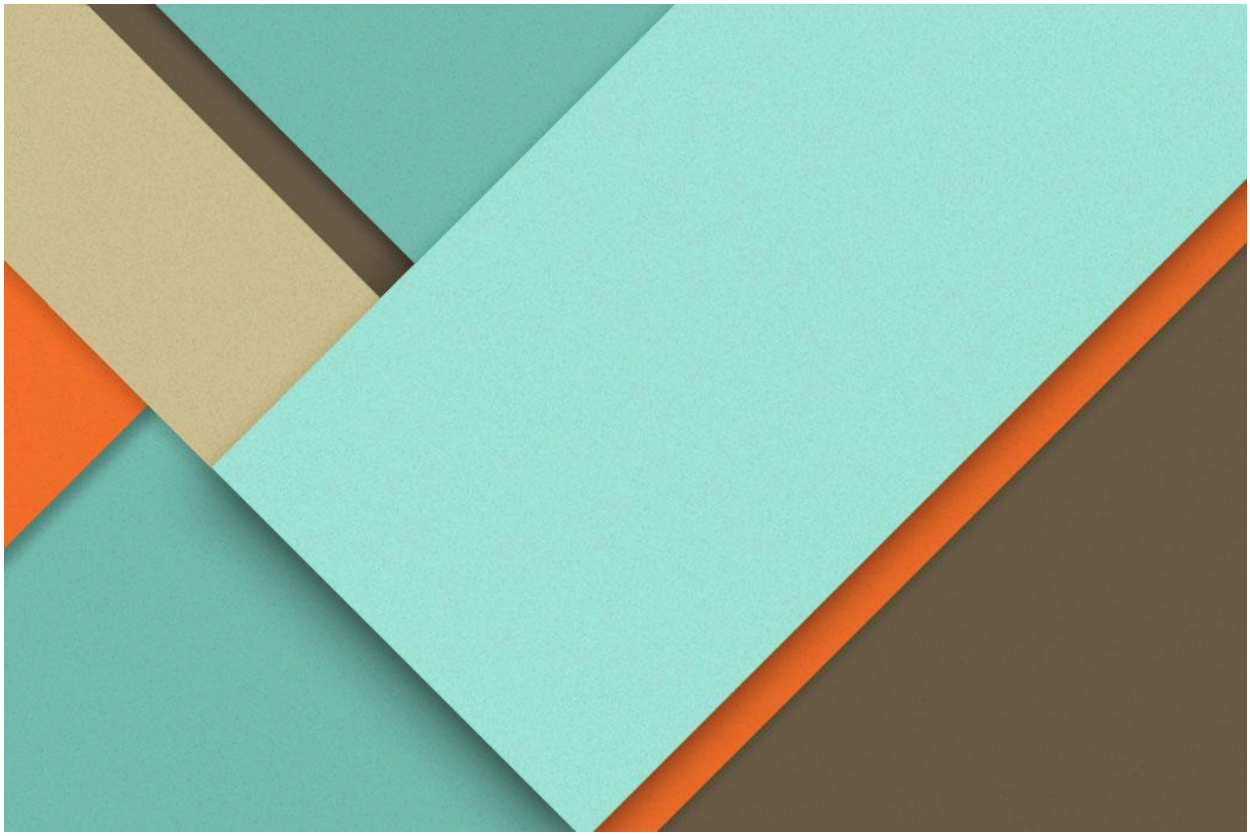

Mega Tasks and Assignments

02.03.2026

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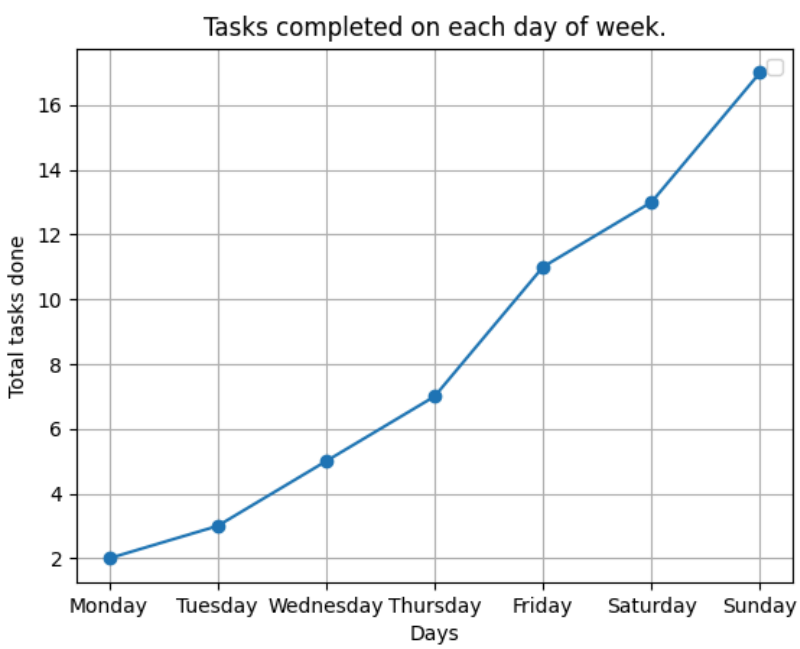
Serial number: 20
Python Development Course - PITP

Mega Task 1

```
#Create a line chart that shows how many tasks have been completed each day of the week. This should represent the progress made over time.

import matplotlib.pyplot as plt


days=["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]

tasks_done=[2, 3, 5, 7, 11, 13, 17]


plt.plot(days,tasks_done,marker='o')

plt.title("Tasks completed on each day of week.")

plt.xlabel("Days")

plt.ylabel("Total tasks done")

plt.grid()

plt.legend()

plt.show()
```

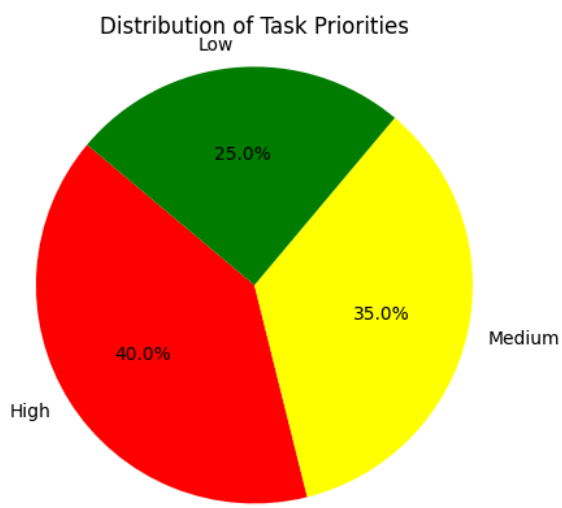
Mega Task 1 Output



Mega Task 2

```
#Create a pie chart that displays the distribution of task priorities (High, Medium, Low).
Ensure that the chart includes percentages.

import matplotlib.pyplot as plt

labels = ['High', 'Medium', 'Low']

sizes = [40, 35, 25]

colors = ['red', 'yellow', 'green']

plt.pie(sizes, labels=labels, colors=colors, autopct='%1.1f%%', startangle=140)

plt.title("Distribution of Task Priorities")

plt.axis('equal')

plt.show()
```

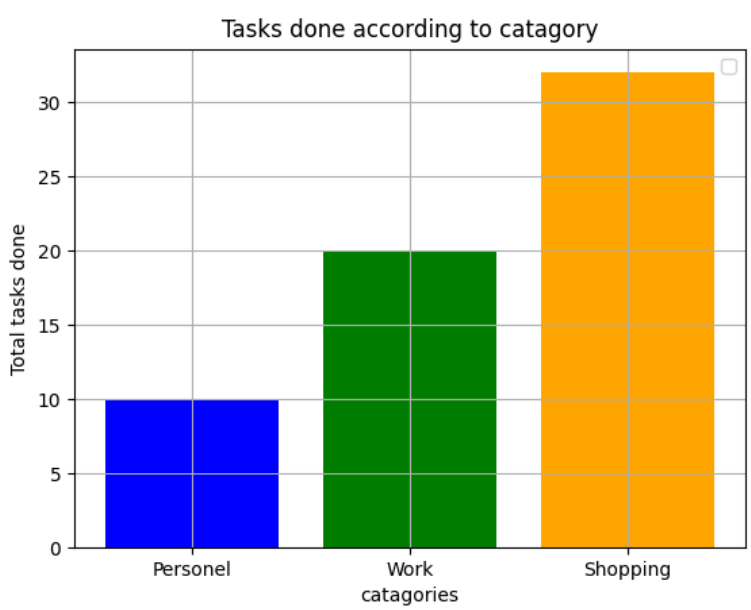
Mega Task 2 Output



Mega Task 3

```
#Using the categories (e.g., Work, Personal, Shopping), create a bar chart to show the task
completion rate for each category.

import matplotlib.pyplot as plt

catagories=["Personel","Work","Shopping"]

tasks_done=[10,20,32]

plt.bar(catagories,tasks_done,color=["blue","green",'orange'])

plt.title("Tasks done according to catagory")

plt.xlabel("catagories")

plt.ylabel("Total tasks done")

plt.grid()

plt.legend()

plt.show()
```

Mega Task 3 Output



Mega Task 4

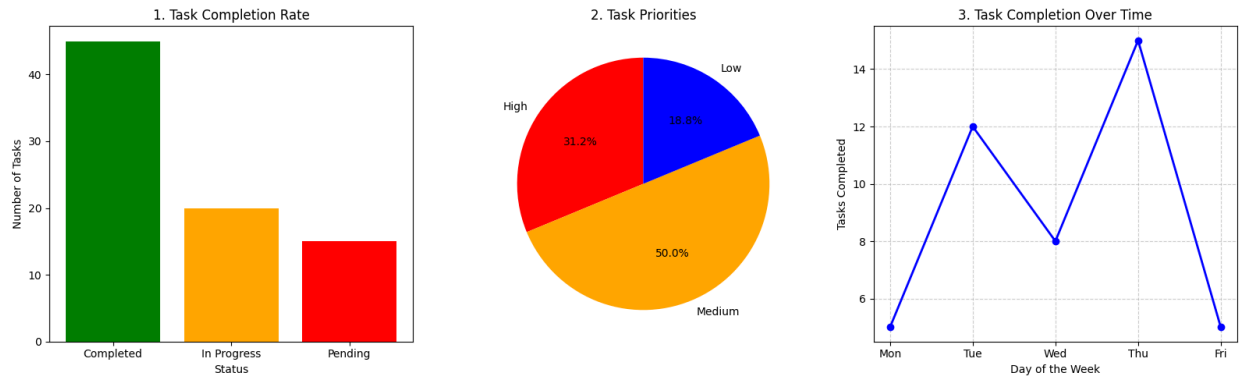
```
#Create a horizontal bar chart that shows how close each task is to its deadline, highlighting
tasks that are approaching their deadlines.

import matplotlib.pyplot as plt

tasks = ['Task A', 'Task B', 'Task C', 'Task D', 'Task E']

days_left = [5, 2, 7, 1, 3]

colors = ['green', 'orange', 'blue', 'red', 'yellow']

plt.barh(tasks,days_left,color=colors)

plt.title("Days left to complete tasks")

plt.grid()

plt.xlabel("Days left")

plt.ylabel("Tasks")

plt.legend()

plt.show()
```

Mega Task 4 Output

Week 7 Assignment 1

```
#Assignment Task:

           #Task Completion Visualization

#Write a Python program that uses `matplotlib` to:

#1. Create a bar chart to visualize the task completion rate.

#2. Use a pie chart to visualize task priorities.

#3. Create a line chart to show task completion over time.

import matplotlib.pyplot as plt

categories = ['Completed', 'In Progress', 'Pending']

task_counts = [45, 20, 15]

priorities = ['High', 'Medium', 'Low']

priority_counts = [25, 40, 15]

colors = ["red","orange","blue"]

days_of_week = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri']

tasks_completed_per_day = [5, 12, 8, 15, 5]

fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(16, 5))

ax1.bar(categories, task_counts, color=['green', 'orange', 'red'])

ax1.set_title("1. Task Completion Rate")

ax1.set_xlabel("Status")

ax1.set_ylabel("Number of Tasks")

ax2.pie(priority_counts, labels=priorities, colors=colors, autopct='%1.1f%%', startangle=90)

ax2.set_title("2. Task Priorities")

ax3.plot(days_of_week, tasks_completed_per_day, marker='o', linestyle='-', color='blue',
linewidth=2)

ax3.set_title("3. Task Completion Over Time")

ax3.set_xlabel("Day of the Week")

ax3.set_ylabel("Tasks Completed")

ax3.grid(True, linestyle='--', alpha=0.6)


plt.tight_layout()

plt.show()
```

Week 7 Assignment 1 Output

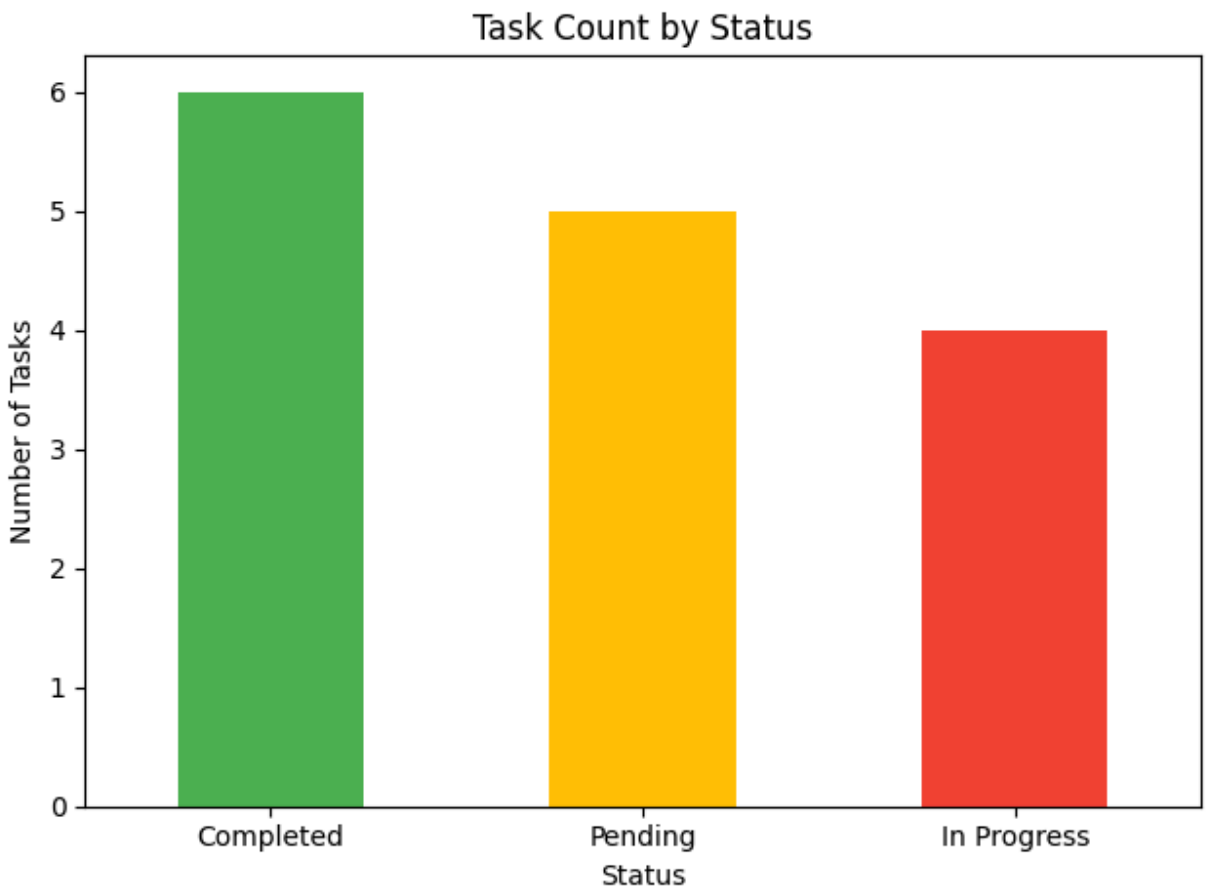


Week 7 Assignment 2

```
#Assignment Task:
#            Task Data Analysis
#Write a Python program that uses `pandas` to:
#1. Read tasks from a CSV file into a DataFrame.
#2. Filter the tasks to show only high priority tasks.
#3. Create a bar chart to visualize the number of tasks by their status.
import pandas as pd
import matplotlib.pyplot as plt
tasks_df = pd.read_csv('tasks.csv')
print("All Tasks:")
print(tasks_df)
print("-" * 30)
high_prio_df = tasks_df[tasks_df['Priority'] == 'High']
print("High Priority Tasks:")
print(high_prio_df)
status_counts = tasks_df['Status'].value_counts()


status_counts.plot(kind='bar', color=["red", "orange", "green"])
plt.title('Task Count by Status')
```

```
plt.xlabel('Status')
plt.ylabel('Number of Tasks')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

Week 7 Assignment 2 Output



Week 8 Assignment 1

Github link:
https://github.com/Ilyan321/Python-Learning/blob/main/week%208/Assignment1.py

Week 8 Assignment 1 Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    GITLENS    AZURE

└─$ python3 -u "/home/ilyan/Study Material/Python-Learning/week 8/Assignment1.py"
1. Add Task
2. View Tasks
3. Mark Task as Complete
4. Search Task
5. Save and Exit
Enter your choice:
```